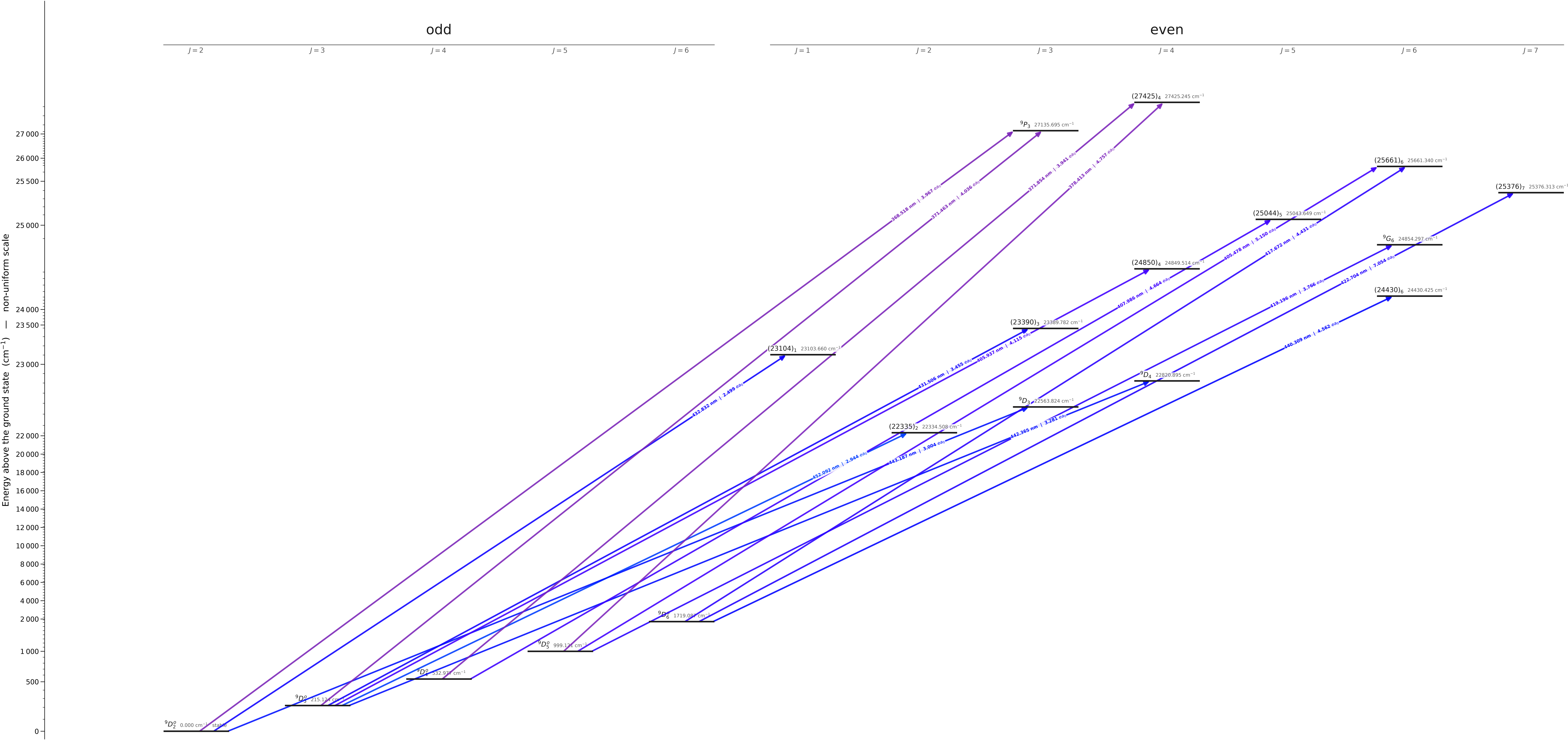
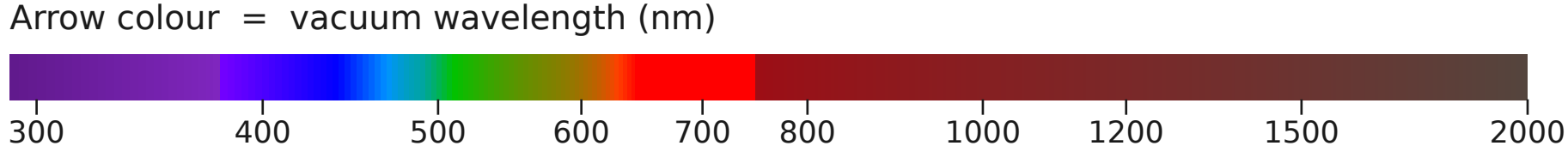


Gadolinium

Gd (Gd I). Strongest classified transitions below 2 μm (NIST): 18 levels, 16 lines.

Gd⁺ ionisation limit 49601.45 cm^{−1} = 6.14980 eV (201.607 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| er \| J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
⁹ D ₂ ^o – ⁹ P ₃	368.5183	368.4134	813.5077	3.967	9.100e+07	–	–	–
⁹ D ₃ ^o – ⁹ P ₃	371.4631	371.3575	807.0584	4.036	9.200e+07	–	–	–
⁹ D ₄ ^o – (27425) ₄	371.8541	371.7483	806.2099	3.941	6.800e+07	–	–	–
⁹ D ₅ ^o – (27425) ₄	378.4134	378.3060	792.2353	4.757	9.400e+07	–	–	–
⁹ D ₆ ^o – (25661) ₆	405.4785	405.3640	739.3547	5.150	6.200e+07	–	–	–
⁹ D ₃ ^o – (24850) ₄	405.9366	405.8220	738.5204	4.115	5.700e+07	–	–	–
⁹ D ₄ ^o – (25044) ₅	407.9856	407.8704	734.8115	4.664	5.900e+07	–	–	–
⁹ D ₅ ^o – (25661) ₆	417.6716	417.5539	717.7707	4.431	4.200e+07	–	–	–
⁹ D ₆ ^o – ⁹ G ₆	419.1962	419.0781	715.1602	3.766	3.000e+07	–	–	–
⁹ D ₆ ^o – (25376) ₇	422.7038	422.5848	709.2258	7.054	8.900e+07	–	–	–
⁹ D ₅ ^o – (23390) ₃	431.5058	431.3845	694.7588	3.455	4.300e+07	–	–	–
⁹ D ₇ ^o – (23104) ₁	432.8319	432.7102	692.6303	2.499	5.200e+07	–	–	–
⁹ D ₆ ^o – (24430) ₆	440.3087	440.1851	680.8688	4.562	3.800e+07	–	–	–
⁹ D ₃ ^o – ⁹ D ₄	442.3649	442.2407	677.7040	3.281	2.800e+07	–	–	–
⁹ D ₅ ^o – ⁹ D ₃	443.1873	443.0629	676.4464	3.004	3.000e+07	–	–	–
⁹ D ₅ ^o – (22335) ₂	452.0922	451.9654	663.1224	2.944	3.800e+07	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).